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Detection and Characterization of Individual Nanoparticles in a Liquid by Photothermal Optical Diffraction and Nanofluidics

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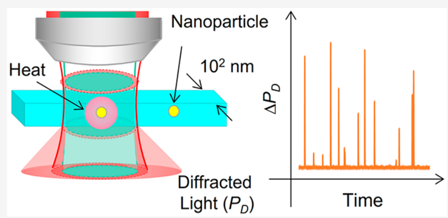
ABSTRACT: Detection and characterization of individual nanoparticles less than 100 nm are important for semiconductor manufacturing, environmental monitoring, biomedical diagnostics, and drug delivery. Photothermal spectroscopy is a light absorptiometry and promising method for detection and characterization because of its high sensitivity and selectivity compared with light scattering or electrical detection methods. However, the characterization of individual nanoparticles in liquids is still challenging for conventional photothermal detection methods. Here, we report a method for the ultrasensitive detection and accurate characterization of individual nanoparticles in liquids by photothermal optical diffraction, which utilizes enhancement of optical diffraction by a nanochannel after light absorption and heat generation of individual nanoparticles in the channel. Our method realized individual 20 nm Au nanoparticle detection with almost 100% detection efficiency by utilizing nanochannels, leading to concentration determination without a calibration curve. Furthermore, we measured individual nanoparticle size and discriminated 20 and 40 nm Au nanoparticles from their photothermal signals. Our photothermal-based nanoparticle detection method in nanochannels has a potential for a wide range of applications such as on-site evaluation of synthesized plasmonic nanoparticles and drug delivery particles.



Nanoparticles in liquids with sizes of less than 100 nm are becoming increasingly important in various fields such as semiconductor manufacturing,[1] environmental monitoring,[2,3] biomedical diagnostics,[4,5] nanomedicine,[6] and photothermal therapy.[7] Consequently, detection and physical/chemical characterization of individual nanoparticles and concentration determination of ultratrace samples are strongly needed. A requirement for detection methods is high sensitivity that is capable of individual nanoparticle detection, real-time and high-throughput detection, and simple and versatile detection systems for a wide range of applications. Several methods for nanoparticle detection and characterization have been developed so far. Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) are often used as direct and accurate measurement methods of particle size distribution,[8] although these methods are time-intensive. As an optical detection method, light-scattering detection methods combined with interferometry,[9] sheath flow cytometry,[10] and tracking analysis[11,12] have realized individual nanoparticle detection and size measurement in liquids. As an electrical detection method, resistive pulse sensing (RPS)[13−15] with nanogaps or nanopores has been used for the detection of individual nanoparticles or viruses in liquids. However, all these methods are lacking in selectivity; it is difficult to distinguish different kinds of nanoparticles with similar size and eliminate signals from impurities such as dust and bubbles.

Absorption-based detection methods overcome these problems because the absorption spectrum is specific to species. Laser-induced fluorescence (LIF)[16,17] is widely used because of the high sensitivity and selectivity; however, it often requires a tedious labeling process. Photothermal spectroscopy (PTS) is widely used as a label-free ultrasensitive absorption-based detection method.[18] The principle of PTS is based on light absorption and heat generation (photothermal effect) by analysis targets. Therefore, it is suitable for the evaluation of nanoparticles synthesized for photothermal therapy[19] and drug delivery.[20] Several photothermal-based nanoparticle detection methods have been reported. For example, thermal lens detection[21] and photothermal differential interference contrast[22] realized single 10 nm nanoparticle detection in micrometer-scale channels. Photothermal microscopy[23] realized imaging of 5 nm Au nanoparticles immobilized on a surface. However, the characterization of individual nanoparticles from their photothermal signals in a liquid is still challenging. In general, photothermal detection methods are highly position-sensitive because they use two focused laser beams. Therefore, signal intensity mainly depends on positions and trajectories of nanoparticles in the optical spots; each peak signal has little information about the size and absorbance of individual nanoparticles. This limitation hinders applying photothermal detection methods to real-time detection and characterization of nanoparticles in a liquid despite the high

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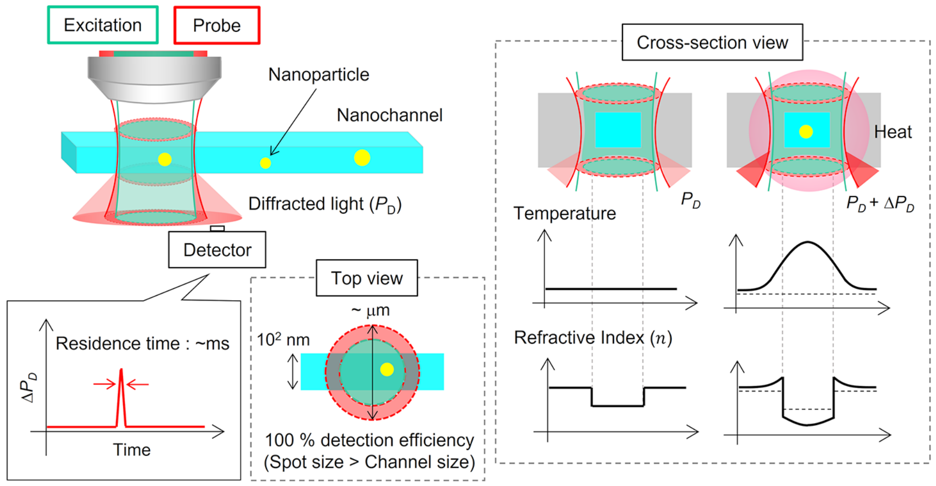

Figure 1. Principle of nanoparticle detection and characterization by counting-mode POD. The right side of the cross-section view represents diffracted light intensity change by the photothermal effect of nanoparticles.

sensitivity and selectivity. As a result, measurements of size and absorbance of individual nanoparticles from photothermal signals have been realized only for nanoparticles immobilized on a surface.[24]

One way to control positions and trajectories of nanoparticles in a liquid is to confine nanoparticles in a nanochannel.[25] The width of the nanochannel (10^1−10^2 nm) is smaller than the focused beam spots (∼μm), and all nanoparticles experience almost the same laser intensity. Therefore, signal intensity depends on the light absorption characteristics of nanoparticles rather than positions and trajectories of nanoparticles. However, for 10^1−10^3 nm channels, the sensitivity of photothermal detection methods rapidly decreases because the effect of thermal diffusion from the channels to the substrates (e.g., glass) becomes dominant.[26] For example, thermal diffusion length in the aqueous solution at 1 kHz modulation is calculated to be ∼7 μm, which is 1 or 2 orders larger than the size of the nanochannels. Then, most of the heat generated by the photothermal effect does not contribute to the signal, and detection performance deteriorates. Thus, it is difficult to apply conventional photothermal detection methods to 10^1−10^2 nm channels.

Recently, we developed a photothermal detection method of nonfluorescent molecules in a 10^2 nm channel: photothermal optical diffraction (POD).[27] Our method realizes the concentration determination of nonfluorescent molecules in a 400 nm channel with a limit of detection of 500 molecules (0.84 zmol). For concentration determination, an average number of molecules in the detection region is recorded as a constant signal, and the time average of the signal during a certain period of time enables highly sensitive measurement with reduced noise. For nanoparticle detection, however, the residence time of individual nanoparticles in the detection region is only a few milliseconds, and each event is recorded as a peak signal. Therefore, the sensitivity is low because of the short excitation time, and noise reduction by a long-time average of the signal is difficult. Thus, it is difficult to apply the conventional POD system to nanoparticle detection and characterization.

In this study, we report individual nanoparticle detection and characterization in a liquid by counting-mode POD. As shown later, our detection method enables the absorbance and size measurement of individual nanoparticles flowing through the 10^2 nm channel. Furthermore, the calculated detection efficiency (number of detected particles/total number of particles) of our method is almost 100%, which enables calibration-less concentration determination of nanoparticle solutions.

EXPERIMENTAL SECTION

Principle of Counting-Mode POD. The principle of nanoparticle detection and characterization by counting-mode POD is shown in Figure 1. The focused probe beam spot (∼μm) is larger than the width of the nanochannel (10^1−10^2 nm), and only a part of the probe beam passes through the channel filled with a sample solution. The propagating speed of light depends on the refractive index of a propagating media; therefore, the refractive index difference between water (n = 1.33) and the substrate made of glass (n = 1.46) gives rise to the relative phase difference of light, which is observed as optical diffraction in the far field. The diffracted light intensity (P_D) depends on the relative phase difference: refractive index difference between glass and water (Δn). Nanoparticles flowing in the channel absorb the excitation beam when passing through the laser focus, followed by nonradiative thermal relaxation and localized temperature change of water and glass (photothermal effect). This temperature change induces refractive index change of water and glass because thermal diffusion length (∼μm for 1 kHz modulation) is larger than the channel size (10^1−10^2 nm), and a part of the generated heat diffuses to the substrate. The refractive index response for temperature (dn/dT) of glass is positive and the dn/dT of water is negative. Therefore, Δn becomes larger by the photothermal effect, which increases P_D. Thus, individual nanoparticles are detected as a diffracted light intensity change (ΔP_D).

The principle of POD enables sensitive nanoparticle detection in 10^2 nm channels because heat diffused to the glass substrate contributes to the signal. For other photothermal detection methods, heat diffusion to the glass substrate leads to thermal loss, which decreases the sensitivity and hinders application to ultrasmall channels. For POD, however, heat diffusion to the glass substrate also increases Δn, leading to an increase of photothermal signals. Thermal diffusion to the glass substrate is difficult to prevent in the ultrasmall space.

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Thus, the principle of POD is essential for nanoparticle detection and characterization in 10^1−10^2 nm channels.

The advantages of nanoparticle detection in 10^2 nm channels by POD are as follows. First, our method can measure the size of individual nanoparticles and distinguish different kinds of nanoparticles from their photothermal signal intensity. In our method, the focused excitation beam spot and confocal length (∼μm) are larger than the width and depth of the nanochannel (10^1−10^2 nm), ensuring that all nanoparticles experience almost the same excitation beam intensity. Furthermore, long thermal diffusion length (∼μm) suppresses the signal variations caused by different nanoparticle trajectories. Thus, the intensity of photothermal signals of individual nanoparticles mainly depends on their absorption cross sections, which are related to the sizes and species. Second, POD enables calibration-less concentration determination by detecting all nanoparticles introduced to the channel. As mentioned above, all nanoparticles introduced to the channel passes through the laser focus, and the detection efficiency (number of detected nanoparticles/total number of introduced nanoparticles) is almost 100%. Therefore, the concentration of the nanoparticle solution is calculated from the number of detected nanoparticles divided by the sample volume introduced to the channel. Thus, the concentration is determined without calibration curve; neither standard samples nor laborious calibration procedures are needed.

Experimental Setup. The experimental setup of counting-mode POD is shown in the Supporting Information, (Figure S1). Details such as focal points of lasers, the diffraction pattern, and the detection point are almost the same as those previously reported.[27] The time constant of the lock-in amplifier was set to 2 ms. The signal intensity was recorded by a data logger for every 2 ms. A pressure-driven flow control system injected sample solutions into nanochannels via microchannels. A pressure controller adjusted the flow rate in the nanochannels. The actual flow rate was checked by the following procedure. First, pure water was injected from the left side of the nanochannels. After the nanochannels were filled with water, an aqueous solution of a nonfluorescent dye, Sunset Yellow FCF (Wako Pure Chemical, Osaka, Japan), was injected from the right side of the nanochannels by the pressure of 100 kPa. The replacement time of the solution in nanochannels was measured by the signal intensity change. The flow rate was calculated from the replacement time and the distance between the detection position and the inlet of the nanochannel. Experimental results are shown in the Supporting Information (Figure S2). The calculated flow rate in the nanochannel was 0.17 mm/s for the pressure of 100 kPa.

Nanofluidic Device. Nanochannels with 6 mm length were fabricated on a fused-silica glass substrate (70 × 30 mm^2, VIOSIL-SQ, Shin-Etsu Chemical Co., Ltd., Japan) by electron-beam lithography and reactive-ion etching. Fabricated channel width and depth are 800 ± 20 and 710 ± 20 nm, respectively. These channel sizes were measured by scanning electron microscopy (SEM) and atomic force microscopy (AFM). For sample injection, U-shaped microchannels were fabricated on another glass substrate by photolithography and reactive-ion etching. Inlet holes were drilled at the end of the microchannels. The width and depth of the microchannels are 500 and 5 μm. Thermal bonding of the two glass substrates connected microchannels and nanochannels.

■ RESULT AND DISCUSSION

Verification of the Detection Principle. First, we confirmed the photothermal signals of nanoparticles in the nanochannel (800 nm wide and 710 nm deep). An aqueous solution of Au nanoparticles with 20 nm in diameter (BBI Solutions, Crumlin, UK) was injected into the channel. The concentration of the solution was 3.6 × 10^{10} particles/mL. The expectation value of the number of molecules in the laser focus was 3.4 × 10^{-2}, much smaller than 1. This value was calculated from the concentration, focused excitation beam spot diameter, and cross-section area of the channel. Figure 2 shows the typical time trace of the signal detected by a photodiode. Peak signals derived from individual nanoparticles were observed above the background signal. These signals were obtained only when both the probe laser and excitation laser irradiated the channel; peak signals were not derived from the light scattering of the probe or excitation beam. Furthermore, average peak signal intensity increased linearly with the excitation beam intensity from 5 to 35 mW. (See the Supporting Information, Figure S3.) These results show that the observed peak signals were derived from the photothermal effect of Au nanoparticles in aqueous solutions.

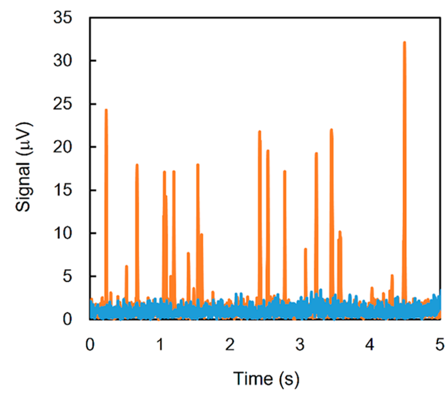

Figure 2. Typical time course signals for 20 nm Au nanoparticle aqueous solution (orange lines). No peak signals were observed without an excitation beam (blue lines).

Furthermore, we confirmed a clear difference in the signal generation mechanism from other photothermal detection methods. We measured the time course signal with a detection position of the slit set at the center of the probe beam spot. This is a general configuration for other photothermal detection methods such as thermal lens detection and photothermal heterodyne detection.[28,29] For POD, however, this configuration is not suitable because the diffraction effect is almost negligible at the center of the probe beam spot. As a result, no peak signals were observed with this configuration. (See Supporting Information, Figure S4.) Therefore, the observed peak signals depend on the optical diffraction; the signal generation mechanism is different from other photothermal detection methods.

Optimization of Experimental Conditions. To realize high sensitivity and detection efficiency, we clarified optimum experimental conditions for counting-mode POD. Figure 3 shows the effect of the flow rate on the number of counted particles and the average peak signal intensity. To count peak signals, we defined a single peak signal as peak signals higher than background signal +5σ with the duration time of more than 15 ms. Symbol σ represents a standard deviation of a

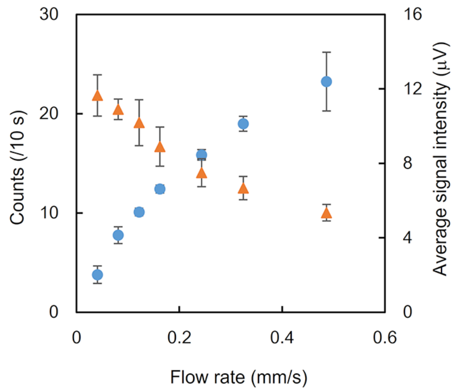

Figure 3. Effect of flow rate on nanoparticle detection. Blue circles represent the number of counts per 10 s. Orange triangles represent the average peak signal intensity.

background signal. Higher flow rate led to high-throughput detection, but the signal intensity decreased because of the short residence time in the detection region, which caused counting loss of nanoparticles. Considering these two factors, the flow rate was set to 0.17 mm/s. At this flow rate, nanoparticles pass through the laser focus within 10 ms. As mentioned in the previous section, higher excitation intensity leads to higher signal intensity. However, we observed trapping of nanoparticles for high excitation intensity. (See Supporting Information, Figure S5.) Therefore, the excitation intensity was set to 30 mW for 20 nm Au nanoparticle detection. Other conditions such as modulation frequency or time constant of a lock-in amplifier were set to 1.1 kHz and 2 ms, which give the highest signal-to-noise ratio for nanoparticle detection.

Concentration Determination of Nanoparticle Solutions. We demonstrated the concentration determination of Au nanoparticle solutions. The number of peak signals per unit time scales linearly if the concentration is low enough to avoid entering two or more nanoparticles in the laser focus at the same time. Figure 4 shows the calibration curve of a 20 nm Au

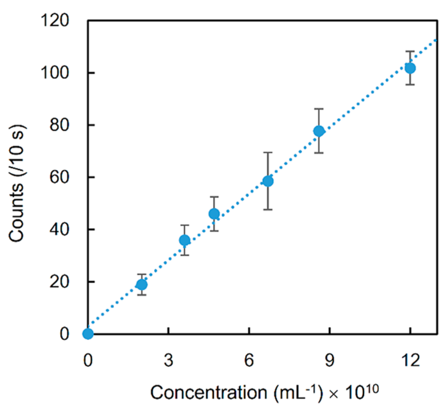

Figure 4. Calibration curve for 20 nm Au nanoparticle aqueous solution.

nanoparticle aqueous solution. The unit time was set to 10 s, and the total measurement time was 5 min for each concentration. The number of peaks per unit time was proportional to the concentration with excellent linearity ($r^2 > 0.99$). One of the advantages of nanochannels for nanoparticle detection is aL−fL detection volume, which enables individual nanoparticle detection, even for highly concentrated samples. For example, the calculated detection volume was 0.92 fL in this experiment, and the expected number of nanoparticles in the detection volume is 4.0×10^{-3}, even for samples with 10^{11} particles/mL concentration. For bulk-scale measurement with the detection volume of mL−nL, individual nanoparticle detection in such highly concentrated samples needs additional procedures such as dilution.

As mentioned in the introduction, our method detects all nanoparticles flowing in the nanochannel, leading to high detection efficiency (number of detected nanoparticles/total number of introduced nanoparticles). Figure 5 shows the

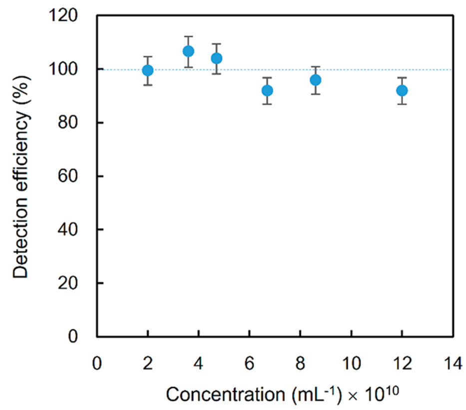

Figure 5. Detection efficiency for each concentration of 20 nm Au nanoparticle aqueous solutions. The dashed line represents 100% detection efficiency.

detection efficiency for each concentration. The total number of nanoparticles introduced into the channel was estimated from the concentration, flow rate, and cross-section area of the channel. Each concentration realized nearly 100% detection efficiency. Lower detection efficiency for higher concentration is explained by miscounting due to an increase in the number of peak signals per unit time.

As a follow-up experiment, we checked the distribution of peak counts per unit time. Each detection event occurs continuously and independently at a constant average rate; therefore, the distribution of the peak counts per unit time should be Poisson distribution. We set unit time to 1 s and counted peak signals for 1000 s. Figure 6 shows the distribution of peak counts per unit time. For comparison, the theoretical Poisson distribution was calculated from the concentration, flow rate, and channel size. The distribution of the peak counts was well-fitted with the Poisson distribution. This result indicates little or no duplicate counting caused by Brownian motion of nanoparticles or optical trapping. Thus, our method accurately detects individual nanoparticles flowing through the laser focus. These results show that counting-mode POD allows for a calibration-less concentration determination by dividing the number of detected nanoparticles by the sample volume introduced into the channel.

Size Measurement of Individual Nanoparticles. Our method ensures that all nanoparticles experience almost the same excitation beam intensity, and each peak signal has information about the physical or chemical properties of individual nanoparticles. Thus, POD can characterize individual nanoparticles in a liquid, which is difficult for other photothermal detection methods. Figure 7 shows the intensity distribution of 5000 peak signals from a 20 nm Au nanoparticle aqueous solution. Note that the bin width of the histogram was

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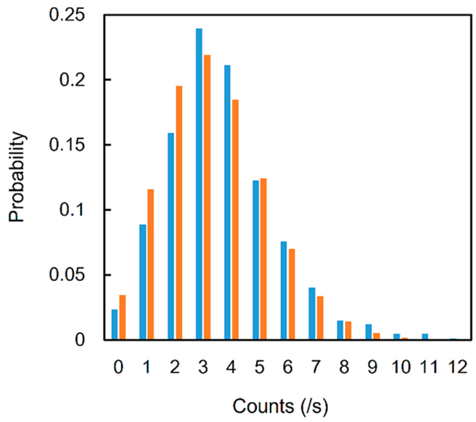

Figure 6. Distribution of peak counts per unit time. The concentration of the solution was 3.6×10^{10} particles/mL. Blue bars represent experimental results. Orange bars represent the expected values calculated from the theoretical Poisson distribution theory.

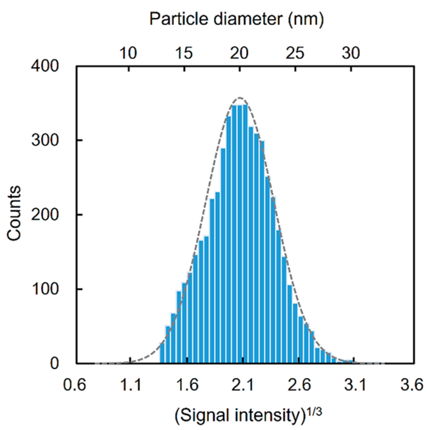

Figure 7. Histogram of the signal intensity for 20 nm Au nanoparticles. Particle diameter was estimated from the average signal intensity.

set on the basis of the cube root of the signal intensity. According to Mie theory, the absorption cross section of metal nanoparticles is proportional to their volume; the photothermal signal is proportional to the cube of the particle diameter. Therefore, the histogram also represents the size distribution of 20 nm Au nanoparticles. The histogram showed unimodal distribution, similar to those obtained from other nanoparticle detection methods.[30,31] The coefficient variation (CV) is 15.1% around the average signal, which corresponds to ±3 nm deviation. The CV is a little larger than those reported by the manufacturer (<8%), which is due to the spatial variation in laser intensity in the channel (~12%). The CV calculated from error propagation theory is <14.6%, almost similar to the experimental value (15.1%). This variation will be suppressed by nanochannels with smaller width and depth. In our method, size distribution was estimated from the absorbance of individual nanoparticles. To the best of our knowledge, no other nanoparticle detection method precisely measured the absorbance of individual nanoparticles in a liquid. Our detection method realized real-time individual nanoparticle detection and characterization with a detection rate of more than 1000 particles/min. Although the throughput is higher than other detection methods requiring surface immobilization of nanoparticles, the flow volume of our method is 5.8 pL/min. Therefore, our method is suitable for the ultrasmall volume of samples such as cells, and long-time measurement or preconcentration is necessary for low-concentration samples.

Finally, our method distinguished nanoparticles with different sizes or absorbance in the same solutions. Figure 8 shows the particle size distribution for a mixture of nanoparticles with diameters of 20 and 40 nm. To avoid optical force trapping, the excitation intensity was set to 20 mW and the probe intensity was set to 8 mW. Two particle sizes were resolved. The particle sizes in the horizontal axis were calculated on the basis of the average signal intensity of pure 20 nm Au nanoparticle aqueous solution, assuming that the signal is proportional to the particle diameter.[3] Two peak positions are well accorded with the sizes of each nanoparticle (20 and 40 nm). By introducing a threshold at $(\text{signal})^{1/3} = 3.0$, we calculated average particle sizes for two peaks as 20.7 ± 3.0 nm (CV = 14.3%) and 40.9 ± 5.4 nm (CV = 13.4%), respectively. From these results, we conclude that our detection method distinguished nanoparticles with different absorbances or sizes in the same solution with high resolution.

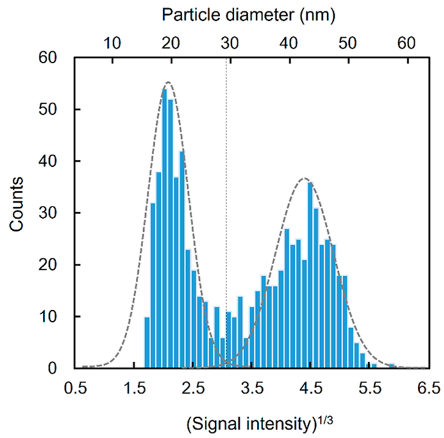

Figure 8. Size distribution for a mixture of 20 and 40 nm Au nanoparticles. The vertical line represents the threshold of two peaks for average-size calculation.

■ CONCLUSION

We have developed a new nanoparticle detection and characterization method by counting-mode photothermal optical diffraction (POD) and nanofluidics. Our method realized calibration-less concentration determination and absorbance measurement of individual nanoparticles in a liquid. The important feature of POD is label-free and real-time detection of individual nanoparticles without the need for surface immobilization, which enables online detection of synthesized nanoparticles on micro-/nanofluidic devices. Also, combining fluidic control systems such as valves will allow for the sorting of individual nanoparticles based on their sizes or compositions. With use of UV laser as an excitation beam, biological samples such as viruses and exosomes can be detected. Different from other detection methods such as light scattering and resistive pulse sensing, POD measures light absorbance and heat generation of individual nanoparticles. This feature enables selective detection of target molecules in real samples containing impurities. Also, drug molecules confined in individual drug delivery carriers with similar sizes

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can be quantified from their absorbance. Furthermore, our method can directly evaluate heat conversion efficiencies of nanoparticles engineered for photothermal therapy and drug delivery. The problem of adsorption and clotting of nanoparticles can be solved by adjusting channel size and modifying the channel surface (e.g., polyethylene glycol). Introducing multiple lasers with different wavelengths and modulation frequencies will further improve the discrimination of nanoparticles with different sizes or types.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.9b05554.

> Signal intensity dependence on excitation intensity, confirmation of signal generation mechanism, typical time course signal of optical force trapping, experimental setup, and flow rate measurement (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. K.M. conceived the project and designed the experiments. Y.T. designed and performed the experiments.

Notes

The authors declare no competing financial interest.

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